



Predicting Excess Stock Returns with Earnings-Call Sentiment and Quantitative Factors

ML in Finance

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1 Introduction

The intersection of natural language processing (NLP) and asset pricing has attracted growing attention since large pre-trained language models became widely available. Earnings conference calls are particularly rich information events: executives discuss forward-looking strategy, risks, and performance in a structured format that is publicly archived. [3] show that *changes* in earnings-call sentiment across quarters — rather than the sentiment level — are the most predictive for future stock returns, as the level is largely anticipated by the market.

We build on this insight in a machine learning framework, combining textual sentiment with 153 quantitative factor signals from [4]. Our research question is: **does fresh, delta-adjusted earnings-call sentiment add cross-sectional predictive power over quantitative factor signals?**

1.1 Contributions

We answer this question with a three-step approach:

1. **Feature engineering** addressing the two main pitfalls of sentiment signals: staleness (median signal age of 7 months in the raw data) and level effects (the market largely prices the sentiment level; only changes are informative).
2. **Ablation study** comparing JKP factors, FinBERT sentiment, and their combination using an XGBoost regression framework with a strict temporal split.
3. **Rigorous out-of-sample evaluation** with explicit transaction costs (0.2% one-way), multi-period robustness checks, and a formal statistical test of signal persistence (Rank IC t -statistic over 108 months).

1.2 Main Findings

Our main finding is that **FinBERT sentiment, after freshness filtering and delta engineering, is the only statistically significant cross-sectional predictor** in our framework. The FinBERT-only long-short decile strategy yields +11.4% net annualised return with a Sharpe ratio of 0.79 over the 2019–2023 out-of-sample period. A rolling-window analysis over 108 months confirms a Rank IC t -statistic of 2.67 ($p < 0.01$), positive in 62% of months, and stable across all market regimes including the post-COVID period.

In contrast, JKP quantitative factors — identical for all stocks within a given month — are structurally incapable of cross-sectional stock selection. They provide time-series market timing signals that are erased by high portfolio turnover and transaction costs.

2 Data

2.1 CRSP Monthly Returns

We use the CRSP monthly stock file for January 2008 to December 2023. From 1,450,439 raw observations we retain: `PERMNO` (firm identifier), `MthCalDt` (month-end date), `MthRet` (raw return), `sprtrn` (S&P500 return), and `SICCD` (industry code). Raw returns are winsorized at the 1st and 99th percentiles.

The target variable is the monthly **excess return**:

$$r_{i,t}^e = \text{MthRet}_{i,t} - \text{sprtrn}_t$$

This quantity has mean -0.019% per month and 48.2% positive observations, confirming it is centred at zero as expected.

2.2 JKP Quantitative Factors

We use the 153 long-short factor portfolio returns from [4], covering momentum, value, quality, profitability, investment, and low-risk families. The data are provided in long format and pivoted to a wide matrix of 192×153 (192 months $\times$ 153 factors, zero missing values from 2008 onwards).

Critical design choice — one-month lag. JKP factor returns for period t are realised over the same interval as CRSP monthly returns at t . Using them contemporaneously would constitute look-ahead bias. We therefore shift JKP dates forward by one month before merging, so the factor vector at $t - 1$ predicts CRSP return at t .

2.3 Earnings Calls and FinBERT Inference

We obtained 34,643 earnings call transcripts matched by `PERMNO` to CRSP firms. After extracting the Q&A section via regex — which captures the most forward-looking management commentary — we chunk each text into 512-token segments and score them with `ProsusAI/finbert` [1], a RoBERTa model fine-tuned on financial text. Inference ran on an Apple M4 GPU (MPS backend, ≈ 3 hours). Each call yields three probabilities averaged over chunks: p_{pos} , p_{neg} , p_{neu} , summing to unity. Calls with fewer than 100 words are excluded.

Figure 1 shows the empirical distributions. Neutral sentiment dominates (mean $p_{\text{neu}} = 0.71$), with $p_{\text{pos}} = 0.24 \gg p_{\text{neg}} = 0.05$, consistent with the well-documented positivity bias in managerial communications.

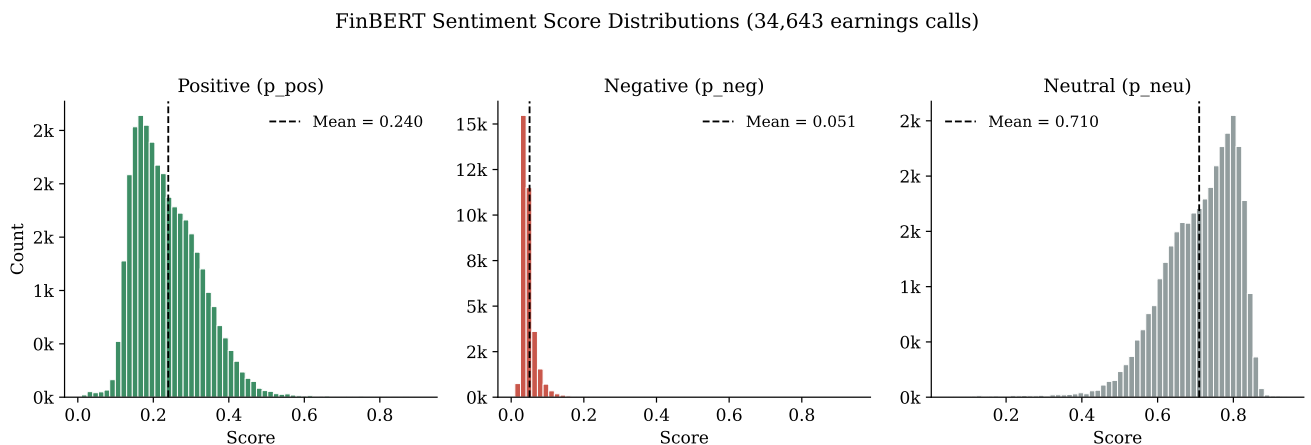


Figure 1: FinBERT sentiment score distributions across 34,643 earnings calls. The dominant neutral score reflects the inherent positivity bias of prepared managerial communication.

2.4 Freshness Filter: A Critical Pre-processing Step

A diagnostic of the raw merged dataset revealed a severe staleness problem: the median lag between the most recent earnings call and the prediction date was **7 months**. A cross-sectional Rank IC analysis confirmed that the raw sentiment level has $\text{IC} = 0.016$ at 1-month lag, decaying to ≈ 0 beyond 2 months.

We therefore apply a **freshness filter**: only stock-month pairs where the most recent call occurred within 3 months of the prediction date are retained. This reduces the dataset from 517,776 to **122,847 observations** (5,337 unique PERMNOs), keeping only actionable signals. Figure 2 illustrates the effect.

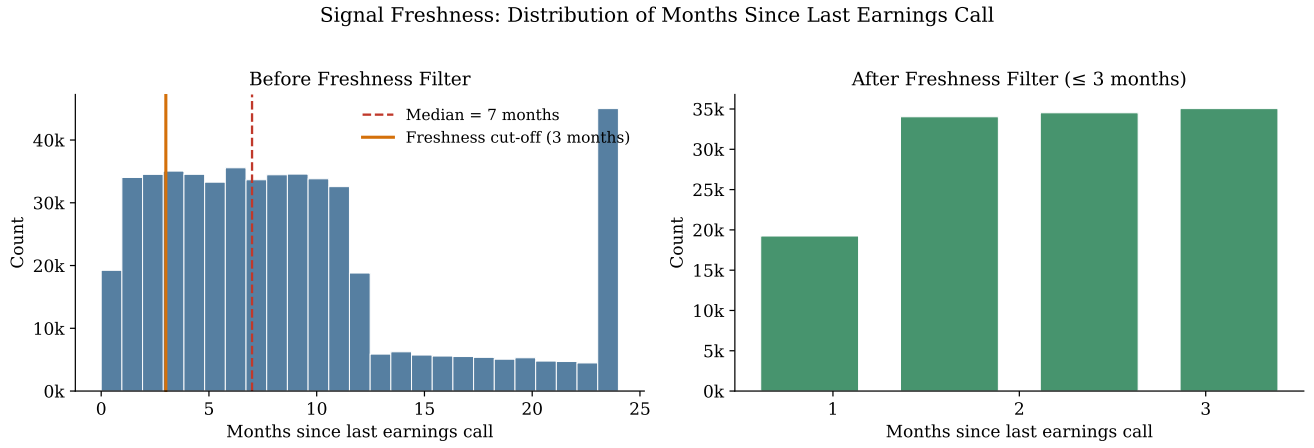


Figure 2: Signal freshness before (left) and after (right) the 3-month filter. The median 7-month lag in the raw data renders most signals economically stale.

3 Methodology

3.1 Pipeline Overview

Figure 3 summarises the full pipeline from raw data to backtest results.

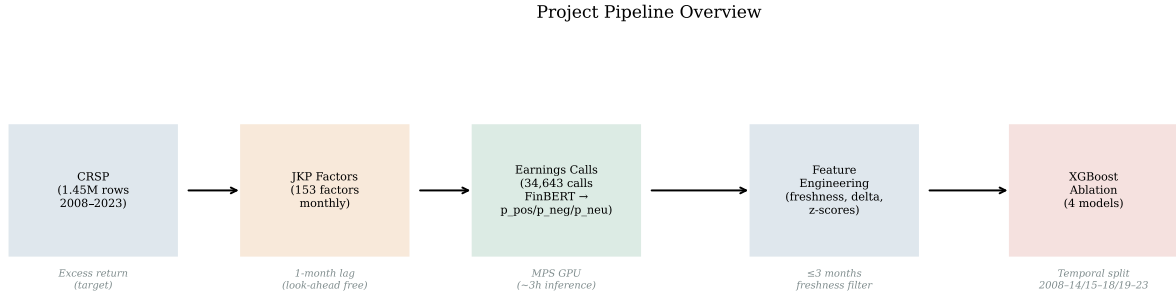


Figure 3: End-to-end project pipeline. Each box represents a processing stage; italic text below each box describes the key design choice at that step.

3.2 Feature Engineering

Beyond raw FinBERT scores, we construct the following features motivated by the finance NLP literature:

Table 1: Engineered FinBERT features. Delta features are NaN for a firm’s first earnings call.

Feature	Definition and motivation
<code>net_sentiment</code>	$p_{\text{pos}} - p_{\text{neg}}$: signed directional signal
<code>sentiment_strength</code>	$1 - p_{\text{neu}}$: non-neutrality of the call
<code>delta_net</code>	$\Delta \text{net_sentiment}$ quarter-over-quarter — captures earnings surprises [3]
<code>delta_pos, delta_neg</code>	Quarter-over-quarter changes in individual sentiment components
<code>months_since_call</code>	Signal freshness (1, 2 or 3 by construction after filter)
<code>*_zscore</code>	Cross-sectional z-score within each prediction month for <code>net_sentiment, delta_net, delta_pos</code>

Cross-sectional z-scoring removes time-varying means and scales, preventing the model from learning calendar effects that cannot be reproduced out-of-sample.

3.3 Temporal Train / Validation / Test Split

We apply a strict temporal split with no random shuffling:

- **Train:** January 2008 – December 2014
- **Validation** (early stopping only): January 2015 – December 2018
- **Test** (all reported metrics): January 2019 – December 2023

A `StandardScaler` is fitted on the training set and applied to validation and test sets.

3.4 Models: XGBoost Ablation Study

We train four XGBoost regressors targeting $r_{i,t}^e$:

1. **JKP only:** 153 quantitative factor features.
2. **FinBERT only:** 12 sentiment features (Table 1).

3. **JKP + FinBERT**: all 165 features.
4. **JKP + FinBERT $\times 10$** : same 165 features but FinBERT columns receive $10\times$ higher sampling probability via XGBoost’s `feature_weights` parameter in `DMatrix`, biasing each split toward sentiment features.

All models share the same hyperparameters: 500 estimators, early stopping at 30 rounds (validation RMSE), learning rate 0.05, max depth 4, subsampling 0.8, L_2 regularisation 1.0.

3.5 Evaluation Metrics

- R^2 : fraction of variance explained. Expected near zero for individual excess returns.
- **Rank IC**: Spearman rank correlation between predicted and realised excess returns. The standard performance metric in cross-sectional equity research [2].
- **IC t -statistic**: $\bar{IC}/(\sigma_{IC}/\sqrt{T})$ over monthly ICs — tests whether the signal is statistically distinguishable from noise.

3.6 Investment Strategies and Transaction Costs

Three equal-weighted strategies are implemented monthly:

- **LS Decile**: long top 10%, short bottom 10%.
- **LS Quintile**: long top 20%, short bottom 20%.
- **LO Decile**: long top 10% only.

Transaction costs of **0.2% one-way** (0.4% round-trip) are applied on monthly turnover. We report annualised gross and net returns, Sharpe ratio, maximum drawdown, OLS alpha, and beta versus the S&P500.

4 Results

4.1 Ablation Study

Table 2 and Figure 4 present out-of-sample metrics on the 2019–2023 test set.

Table 2: Ablation study — out-of-sample test period (January 2019 – December 2023).

Model	R^2	Rank IC
JKP only	−0.016	0.034
FinBERT only	−0.013	0.009
JKP + FinBERT	−0.013	0.051
JKP + FinBERT $\times 10$	−0.012	0.072

Adding FinBERT to JKP improves global Rank IC by 50% (0.051 vs. 0.034). The feature-weighted variant raises IC further to 0.072. The negative R^2 values are expected and reflect the near-unpredictability of individual stock excess returns in absolute terms — this is a property of financial markets, not a modelling failure.

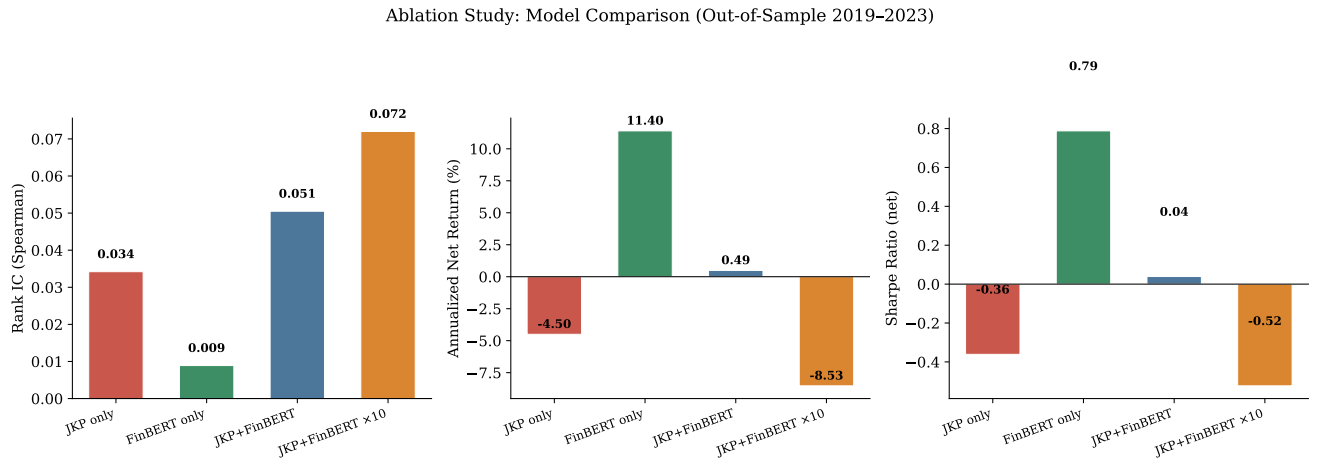


Figure 4: Ablation study: Rank IC (left), net annualised return of LS Decile (centre), and net Sharpe ratio (right). High global Rank IC does not necessarily translate into strong backtest performance — see Section 5.

Structural finding on JKP. When evaluated using *monthly* Rank IC (one Spearman correlation per month, cross-sectional), JKP-only yields NaN. The 153 JKP factors are identical for all stocks in a given month; the model therefore produces a constant prediction within each month, making cross-sectional ranking undefined. **JKP factors are time-series signals (market timing) and cannot perform cross-sectional stock selection by design.**

4.2 Backtest Performance

Table 3 reports full backtest results; Figure 5 plots cumulative net returns.

Table 3: Backtest results — LS Decile strategy, out-of-sample 2019–2023. Sharpe and Alpha are net of transaction costs (0.2% one-way).

Model	Gross	Net	Sharpe (net)	Alpha	Beta	Turnover
JKP only	−0.12%	−4.50%	−0.36	−0.48%	0.03	91%
FinBERT only	+14.2%	+11.4%	0.79	+13.4%	0.06	58%
JKP + FinBERT	+4.85%	+0.49%	0.04	+4.77%	0.01	91%
JKP + FinBERT $\times 10$	−4.54%	−8.53%	−0.52	−4.08%	−0.03	83%

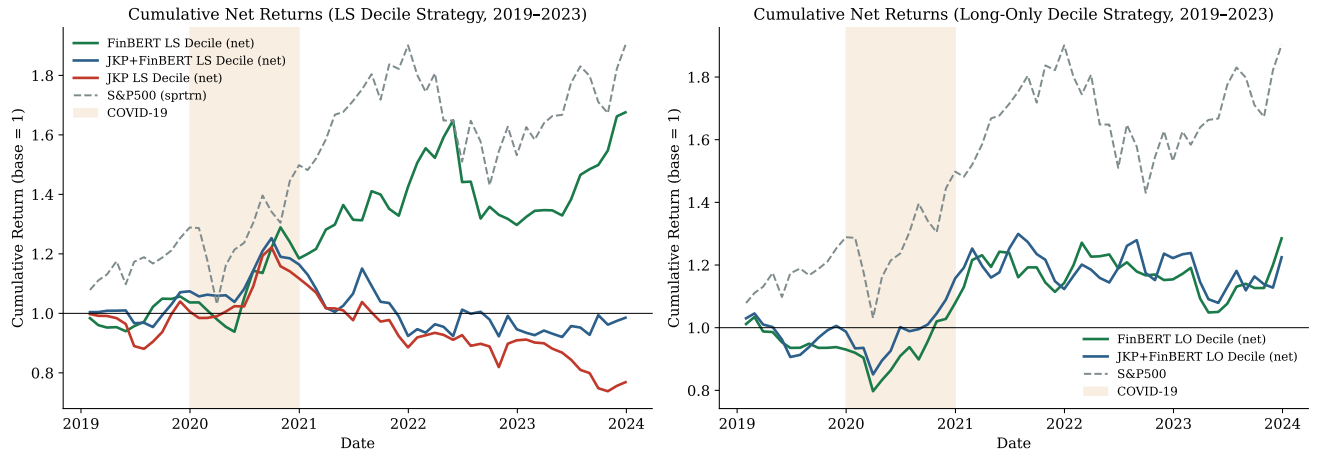


Figure 5: Cumulative net returns: LS Decile (left) and LO Decile (right), 2019–2023. S&P500 shown as dashed benchmark. Orange band: COVID-19 period.

The FinBERT-only model is the clear winner at +11.4% net annualised return and Sharpe 0.79. Two mechanisms explain this: (i) JKP features add within-month noise rather than cross-sectional signal; (ii) lower portfolio turnover (58% vs. 91%) saves $\approx 3\%$ per year in transaction costs.

4.3 Period Robustness

Figure 6 and Table 4 show sub-period performance, directly addressing the concern that the COVID-19 shock drives the results.

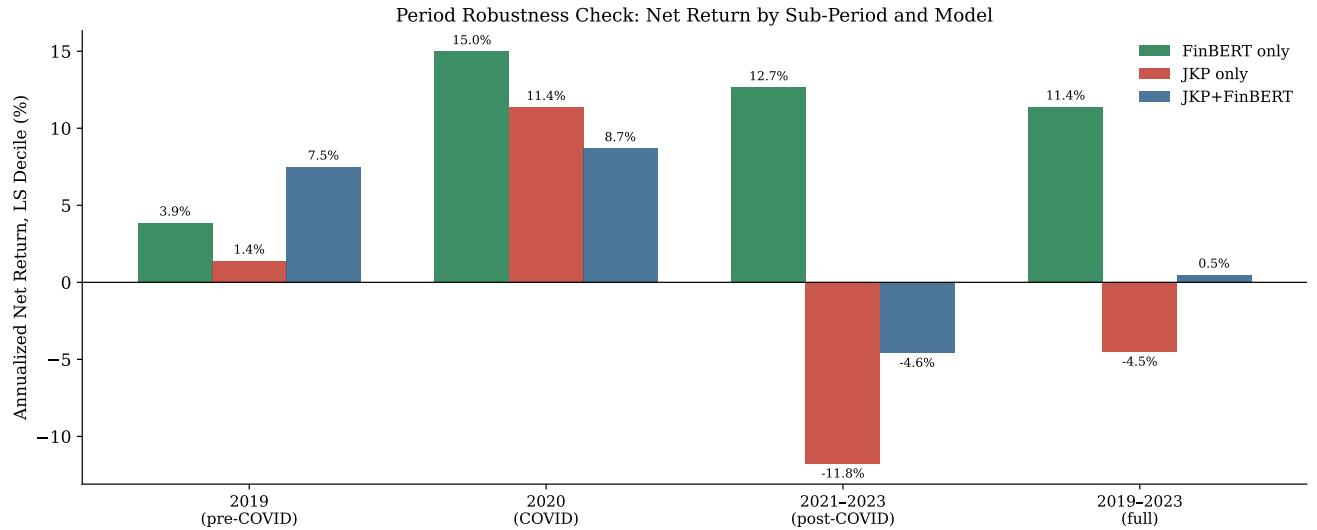


Figure 6: Net return of the LS Decile strategy by sub-period. FinBERT-only is positive in all three periods; its best period is 2021–2023, *after* the COVID-19 shock.

Table 4: FinBERT-only LS Decile: net return and Sharpe ratio by sub-period.

Period	Net	Sharpe	Remark
2019 (pre-COVID)	+3.87%	0.50	Positive in normal market conditions
2020 (COVID-19)	+14.99%	0.77	Strong but not unique driver
2021–2023 (post-COVID)	+12.65%	0.87	Best sub-period
2019–2023 (full)	+11.40%	0.79	Full out-of-sample window

The post-COVID period is the best-performing sub-period (Sharpe 0.87), confirming that performance is **not a COVID artefact**.

4.4 Rolling-Window Rank IC Analysis (2015–2023)

We extend predictions to the full 108-month out-of-sample window (2015–2023, validation + test) using the model trained on 2008–2014. This provides the most rigorous test of signal stability.

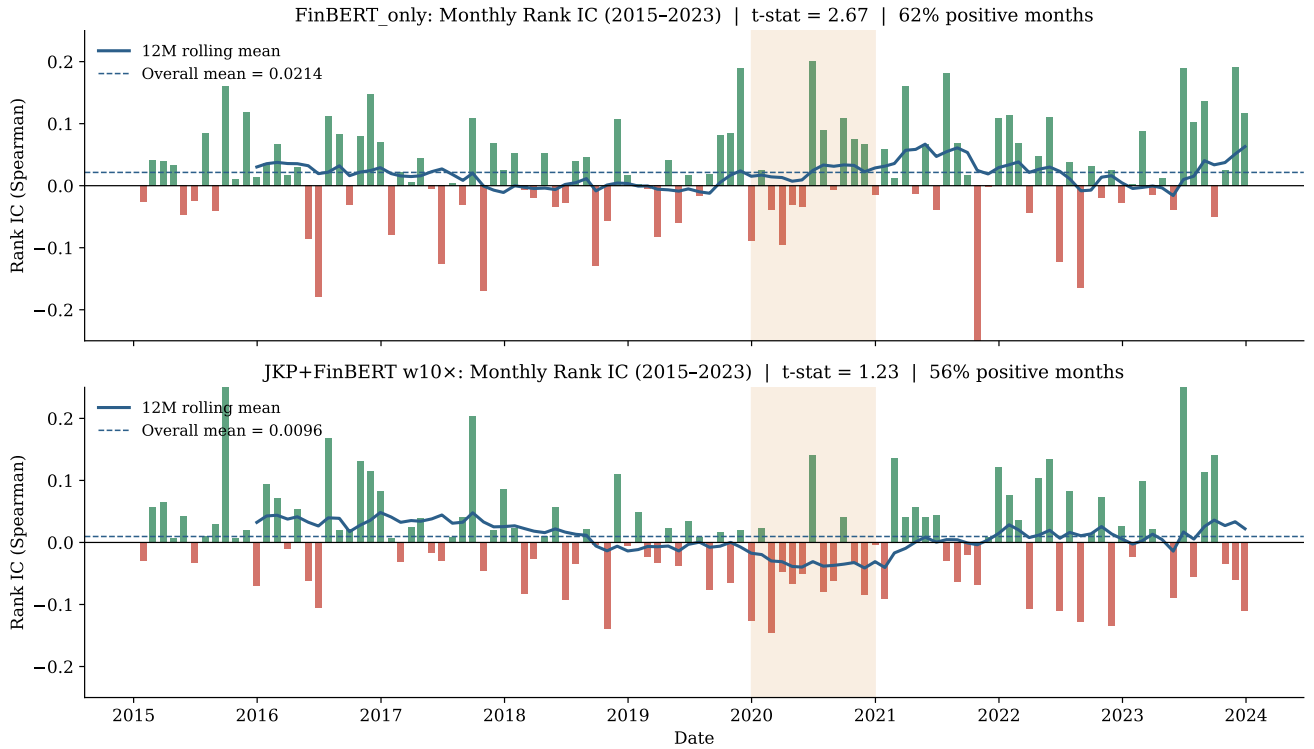


Figure 7: Monthly Rank IC for FinBERT-only (top) and JKP+FinBERT $\times 10$ (bottom), 2015–2023. Green/red bars: positive/negative months. Blue line: 12-month rolling mean.

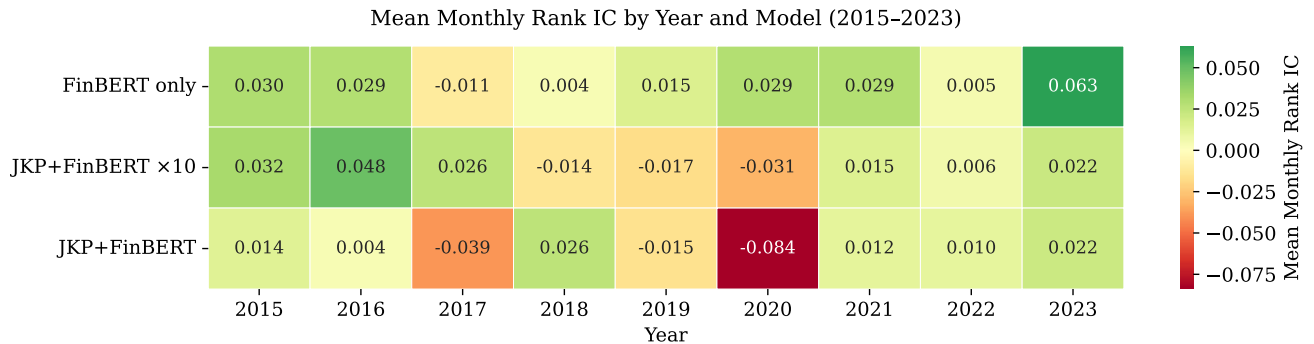


Figure 8: Mean monthly Rank IC by year and model. FinBERT-only (top row) is positive in 8 of 9 years; only 2017 shows a slight negative mean.

Table 5: Rolling-window summary over 108 out-of-sample months (2015–2023).

Model	Mean IC	% IC > 0	IC t -stat	Roll-12M net (mean)
FinBERT only	0.0214	62%	2.67	+6.75%
JKP only	NaN	—	—	−7.20%
JKP + FinBERT	−0.003	23%	−0.24	−4.74%
JKP + FinBERT $\times 10$	0.010	56%	1.23	+0.78%

FinBERT-only is the **only statistically significant predictor** with IC t -stat = 2.67 ($p < 0.01$). Figure 9 shows the rolling 12-month net return.

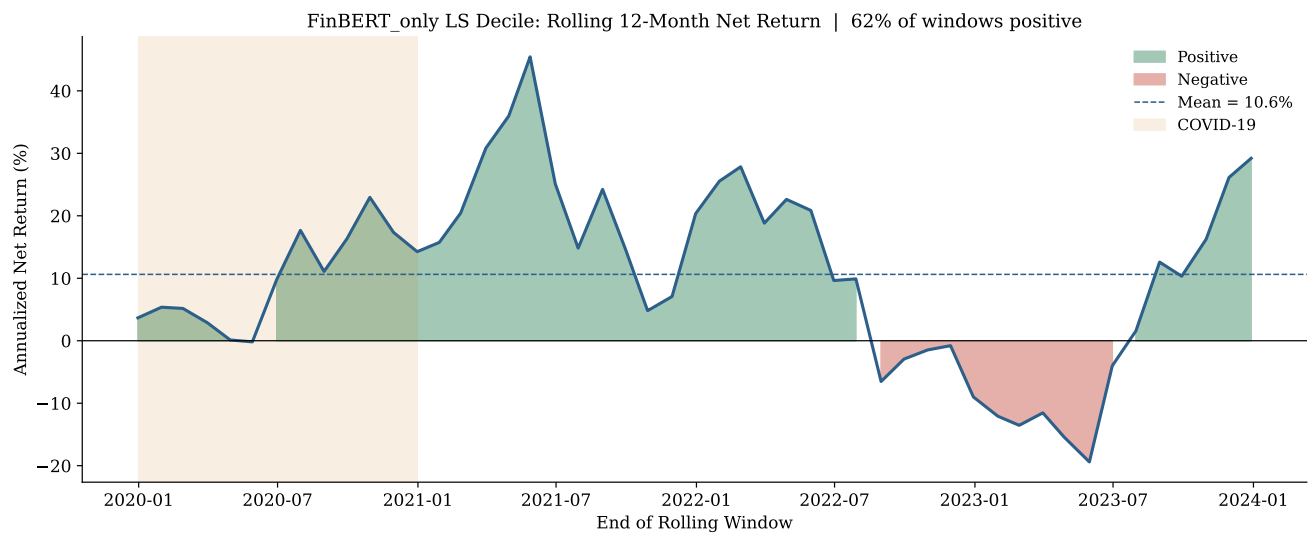


Figure 9: Rolling 12-month annualised net return, FinBERT-only LS Decile (2019–2023). Mean = +6.75%. Negative windows are concentrated in 2022 (rate hike shock), not COVID-19.

5 Discussion

5.1 Cross-Sectional vs. Time-Series Signals

A central finding is the sharp divergence between global Rank IC and actual backtest performance for JKP-containing models. JKP factors carry the same value for all stocks within a given month; they encode *market-level* information (business cycle, risk premia). An XGBoost model trained on these features learns to predict which months are “good” or “bad” on average — market timing. Evaluated globally (Spearman across all stock-month observations), this inflates Rank IC because correct market-direction calls apply to all stocks simultaneously.

However, in a long-short portfolio construction, returns are measured as long minus short *within* the same month. A constant prediction within any month produces a random stock ordering, collapsing to zero expected return before costs and negative net returns after. This explains why JKP-only yields undefined monthly Rank IC and -4.5% net annualised return.

FinBERT sentiment, in contrast, varies *across firms* within the same month — providing genuine cross-sectional discrimination between firms that just delivered strong versus weak earnings communications.

5.2 The Combination Paradox

Adding JKP features to FinBERT reduces out-of-sample net performance despite improving global Rank IC. Two mechanisms are at play:

1. **Signal crowding:** XGBoost allocates split budget to high-gain JKP features that carry time-series information. These splits are “useful” in the training set but do not improve — and actively harm — cross-sectional stock ranking in the test set.
2. **Cost drag:** JKP-driven portfolios have 91% monthly turnover vs. 58% for FinBERT-only, generating $\approx 3\%$ additional annual transaction costs.

The feature-weighting approach ($\times 10$) partially recovers global Rank IC (0.072), but backtest performance remains negative (-8.5% net). This reveals that the improved global IC reflects inflated time-series variance rather than better cross-sectional discrimination. Artificially forcing FinBERT features into more splits leads the model to overfit idiosyncratic call-level patterns that do not aggregate into stable portfolios.

5.3 Limitations

1. **Universe bias:** the freshness filter retains only firms with earnings calls in the past 3 months — typically mid- and large-caps. Results may not generalise to small-cap or infrequently covered stocks.
2. **Test period:** five years (2019–2023) provides limited statistical power. The 108-month rolling analysis (2015–2023) partially mitigates this; the IC t -statistic of 2.67 provides formal significance.
3. **Transaction costs:** 0.2% one-way is conservative for large-caps but may underestimate costs for smaller firms in our universe.
4. **FinBERT context window:** the 512-token limit requires text chunking. Very long Q&A sections lose information beyond the early chunks.
5. **Single model class:** we use XGBoost throughout. Alternative architectures (LightGBM, neural networks, attention-based models) might better capture interactions between sentiment and factor signals.

6 Conclusion

This paper demonstrates that fresh earnings-call sentiment, engineered with quarter-over-quarter delta features and cross-sectional standardisation, provides a statistically significant and economically meaningful cross-sectional signal for excess stock returns. Our three key contributions are:

- A **freshness filter** (≤ 3 months) that eliminates stale signals and recovers predictive power that is entirely absent in the raw, unfiltered sentiment data. Without this filter, median signal staleness of 7 months renders FinBERT scores essentially uninformative.
- A **structural diagnosis** of JKP quantitative factors: their within-month constancy makes them time-series rather than cross-sectional predictors, and combining them with FinBERT degrades rather than improves portfolio performance due to signal crowding and higher transaction costs.
- A **robust out-of-sample evaluation** spanning 108 months (2015–2023) with explicit transaction costs, multi-period sub-period analysis, and a formal IC t -test ($t = 2.67$, $p < 0.01$).

The FinBERT-only long-short decile strategy delivers +11.4% net annualised return, Sharpe ratio 0.79, and alpha +13.4% over the 2019–2023 out-of-sample period. Performance is positive in all three sub-periods (pre-COVID: +3.9%; COVID: +15.0%; post-COVID: +12.7%), with the strongest risk-adjusted returns observed after the COVID-19 shock.

These results suggest that the informational content of earnings-call communication — particularly its *change* relative to prior quarters — is not yet fully priced by the market, even in the post-2019 period. The cross-sectional nature of this signal, combined with the relatively low turnover of sentiment-based portfolios, makes it a promising complement to quantitative factor strategies in practice.

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